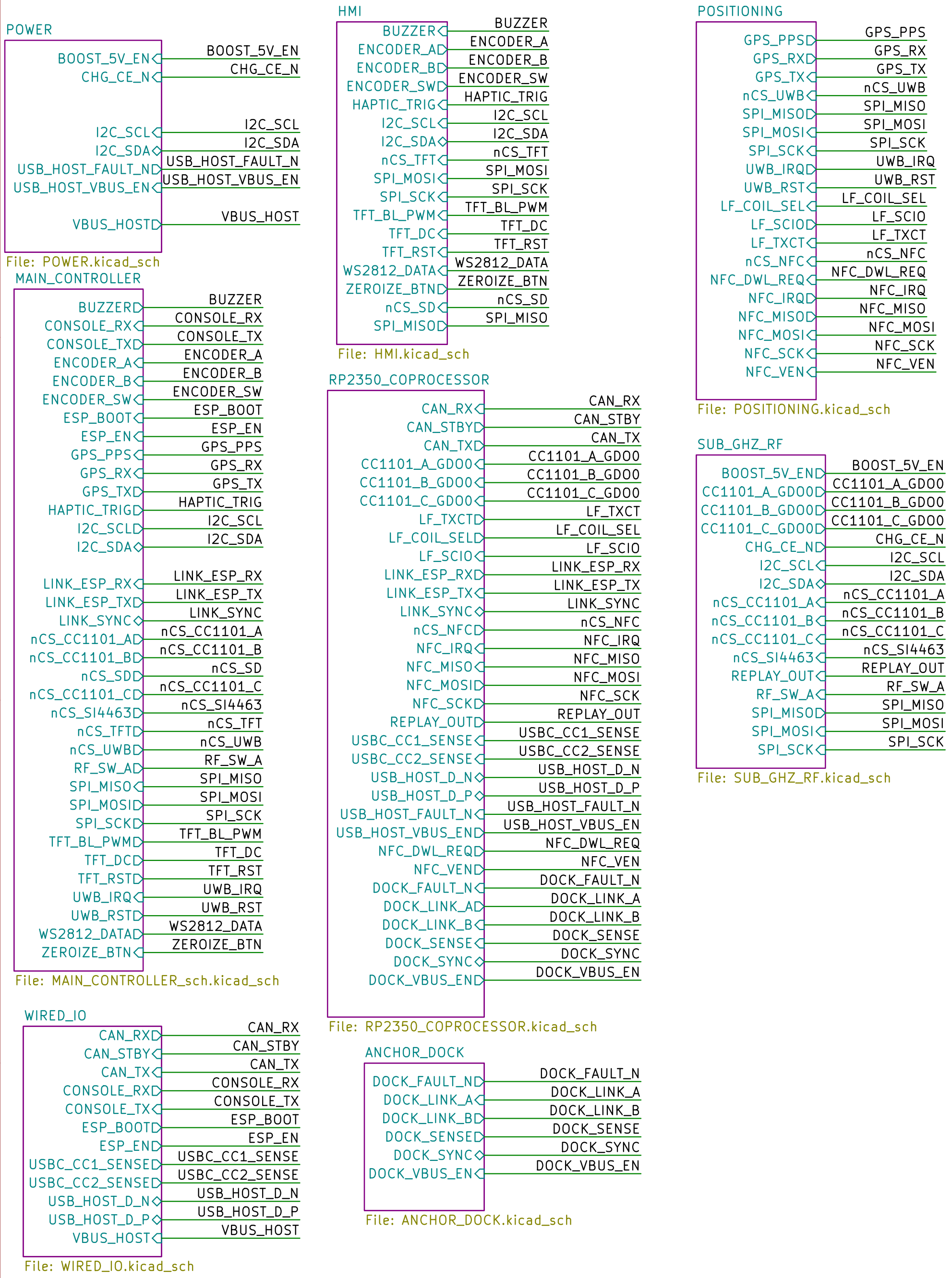


Notes

Power:
Keep CHG_CE_N high, BOOST_5V_EN low and USB_HOST_VBUS_EN low during initialization.
Set charger input-current limit from the detected USB-C source.
Enable +5V_SYS before VBUS_HOST and reverse the sequence during shutdown.

RP2350/WIRED_IO:
Default USB_HOST_VBUS_EN low.
Monitor USB_HOST_FAULT_N and immediately disable VBUS_HOST on an asserted fault.

HMI:
TCA9534 address = 0x21. Configure P0-P3 as inputs before enabling P4 as the common active-low LED sink.
Configure DRV2605L for LRA operation and calibrate for the C08-005 actuator.

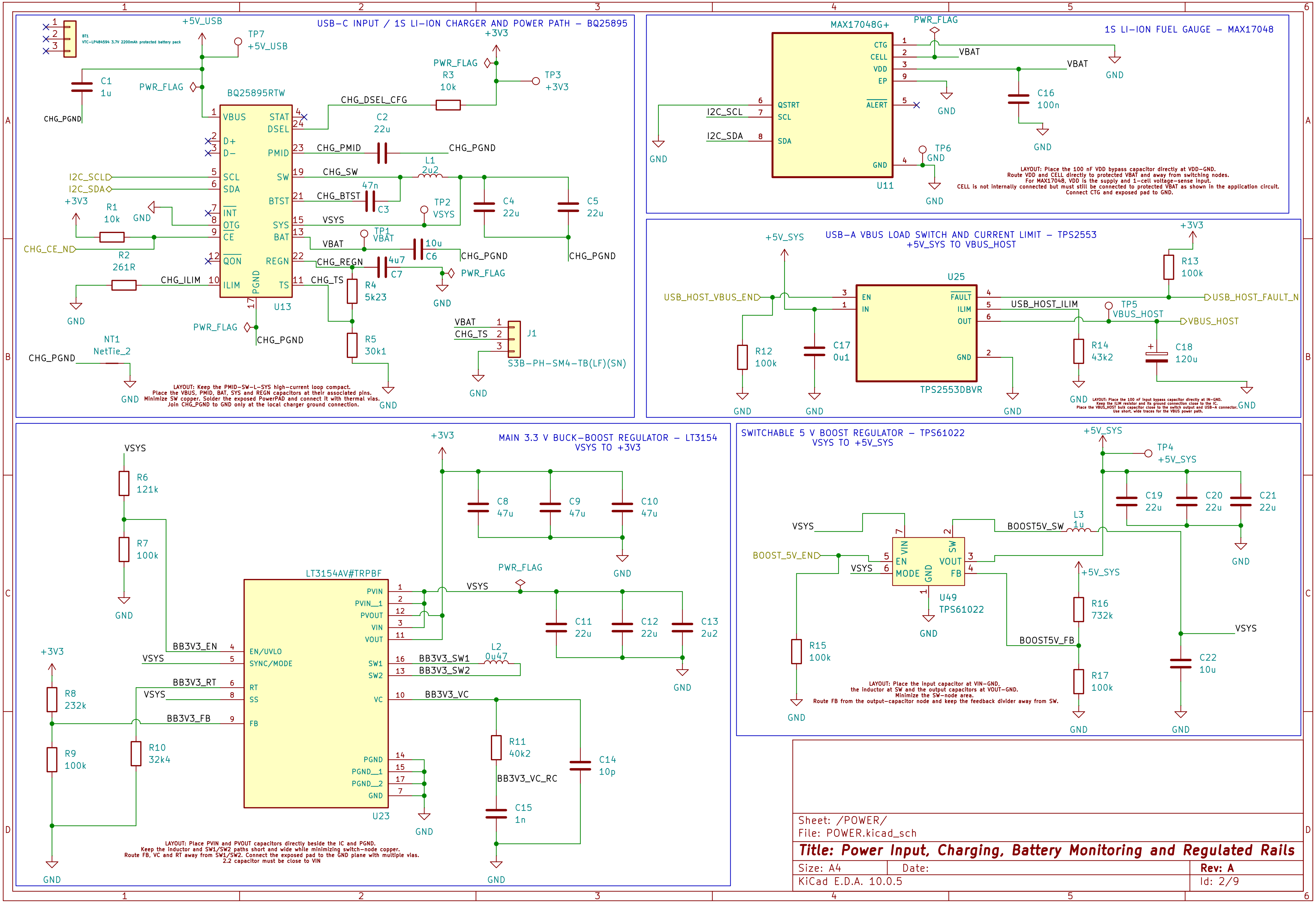
POSITIONING:
MAX-M10N V_BCKP is not powered; expect a cold start after complete power loss.
Hold NFC_VEN and NFC_DWL_REQ low during startup.
Change LF coil selection only while LF transmission is disabled.

CAN assembly:
Populate split termination Resistors only when this PCB is located at a physical end of the CAN bus.

Sheet: /
File: FOBtool_HW.kicad_sch

Title: FOBtool Handheld RF Security Platform

Size: A4	Date:	Rev: A
KiCad E.D.A. 10.0.5		Id: 1/9



USB-C INPUT / 1S LI-ION CHARGER AND POWER PATH – BQ25895
BT1
VTC-LP484594 3.7V 2200mAh protected battery pack
+5V_USB
TP7
+5V_USB
C1
1u
CHG_PGND
PWR_FLAG
BQ25895RTW
U13
VBUS
D+
D–
SCL
SDA
INT
OTG
CE
QON
ILIM
PGND
STAT
DSEL
PMID
SW
BTST
SYS
BAT
REGN
TS
I2C_SCL
I2C_SDA
+3V3
R1
10k
GND
CHG_CE_N
R2
261R
CHG_ILIM
CHG_DSEL_CFG
R3
10k
TP3
+3V3
C2
22u
CHG_PMID
CHG_SW
L1
2u2
CHG_BTST
47n
C3
TP2
VSYS
TP1
VBAT
10u
C6
C4
22u
C5
22u
CHG_REGN
4u7
C7
CHG_TS
R4
5k23
R5
30k1
J1
S3B-PH-SM4-TB(LF)(SN)
VBAT
CHG_TS
NT1
NetTie_2
LAYOUT: Keep the PMID–SW–L–SYS high–current loop compact.
Place the VBUS, PMID, BAT, SYS and REGN capacitors at their associated pins.
Minimize SW copper. Solder the exposed PowerPAD and connect it with thermal vias.
Join CHG_PGND to GND only at the local charger ground connection.
1S LI-ION FUEL GAUGE – MAX17048
MAX17048G+
U11
CTG
CELL
VDD
EP
QSTRT
SCL
SDA
ALERT
GND
VBAT
C16
100n
TP6
GND
LAYOUT: Place the 100 nF VDD bypass capacitor directly at VDD–GND.
Route VDD and CELL directly to protected VBAT and away from switching nodes.
For MAX17048, VDD is the supply and 1–cell voltage–sense input.
CELL is not internally connected but must still be connected to protected VBAT as shown in the application circuit.
Connect CTG and exposed pad to GND.
USB-A VBUS LOAD SWITCH AND CURRENT LIMIT – TPS2553
+5V_SYS TO VBUS_HOST
U25
TPS2553DBVR
EN
IN
FAULT
ILIM
OUT
GND
+5V_SYS
USB_HOST_VBUS_EN
USB_HOST_ILIM
USB_HOST_FAULT_N
VBUS_HOST
TP5
VBUS_HOST
R12
100k
C17
0u1
R13
100k
R14
43k2
C18
120u
LAYOUT: Place the 100 nF input bypass capacitor directly at IN–GND.
Keep the ILIM resistor and its ground connection close to the IC.
Place the VBUS_HOST bulk capacitor close to the switch output and USB–A connector.
Use short, wide traces for the VBUS power path.
MAIN 3.3 V BUCK-BOOST REGULATOR – LT3154
VSYS TO +3V3
LT3154AV#TRPBF
U23
PVIN
PVIN__1
PVOUT
VIN
VOUT
EN/UVLO
SYNC/MODE
RT
SS
FB
SW1
SW2
VC
PGND
PGND__1
PGND__2
GND
R6
121k
R7
100k
R8
232k
R9
100k
R10
32k4
BB3V3_EN
BB3V3_RT
BB3V3_FB
BB3V3_SW1
BB3V3_SW2
BB3V3_VC
BB3V3_VC_RC
L2
0u47
C8
47u
C9
47u
C10
47u
C11
22u
C12
22u
C13
2u2
R11
40k2
C14
10p
C15
1n
LAYOUT: Place PVIN and PVOUT capacitors directly beside the IC and PGND.
Keep the inductor and SW1/SW2 paths short and wide while minimizing switch–node copper.
Route FB, VC and RT away from SW1/SW2. Connect the exposed pad to the GND plane with multiple vias.
2.2 capacitor must be close to VIN
SWITCHABLE 5 V BOOST REGULATOR – TPS61022
VSYS TO +5V_SYS
U49
TPS61022
VIN
EN
MODE
GND
SW
VOUT
FB
TP4
+5V_SYS
BOOST_5V_EN
BOOST5V_SW
BOOST5V_FB
L3
1u
C19
22u
C20
22u
C21
22u
C22
10u
R15
100k
R16
732k
R17
100k
LAYOUT: Place the input capacitor at VIN–GND,
the inductor at SW and the output capacitors at VOUT–GND.
Minimize the SW–node area.
Route FB from the output–capacitor node and keep the feedback divider away from SW.
Sheet: /POWER/
File: POWER.kicad_sch
Title: Power Input, Charging, Battery Monitoring and Regulated Rails
Size: A4
Date:
Rev: A
KiCad E.D.A. 10.0.5
Id: 2/9

LAYOUT — Place U1 at the PCB edge with its antenna facing outward. No copper, tracks, vias, planes, or components are permitted inside the antenna keepout on any PCB layer. Keep switching nodes, USB/UART, SPI, battery, shielding, and metal enclosure parts away from the antenna.

ASSEMBLY — FIT ONLY ESP32-S3-WROOM-1-N16R2.
DO NOT SUBSTITUTE N16R8/R16V: octal-PSRAM variants reserve GPIO35-37, which Rev-A uses for SPI_MOSI, SPI_SCK, and SPI_MISO.

PLACEMENT — Place 22uF and 100nF immediately beside U1 pin 2. Place R19/C24 close to EN pin 3 and R18 close to GPIO0 pin 27. Connect pins 1, 40, and EPAD 41 directly to the solid GND plane.

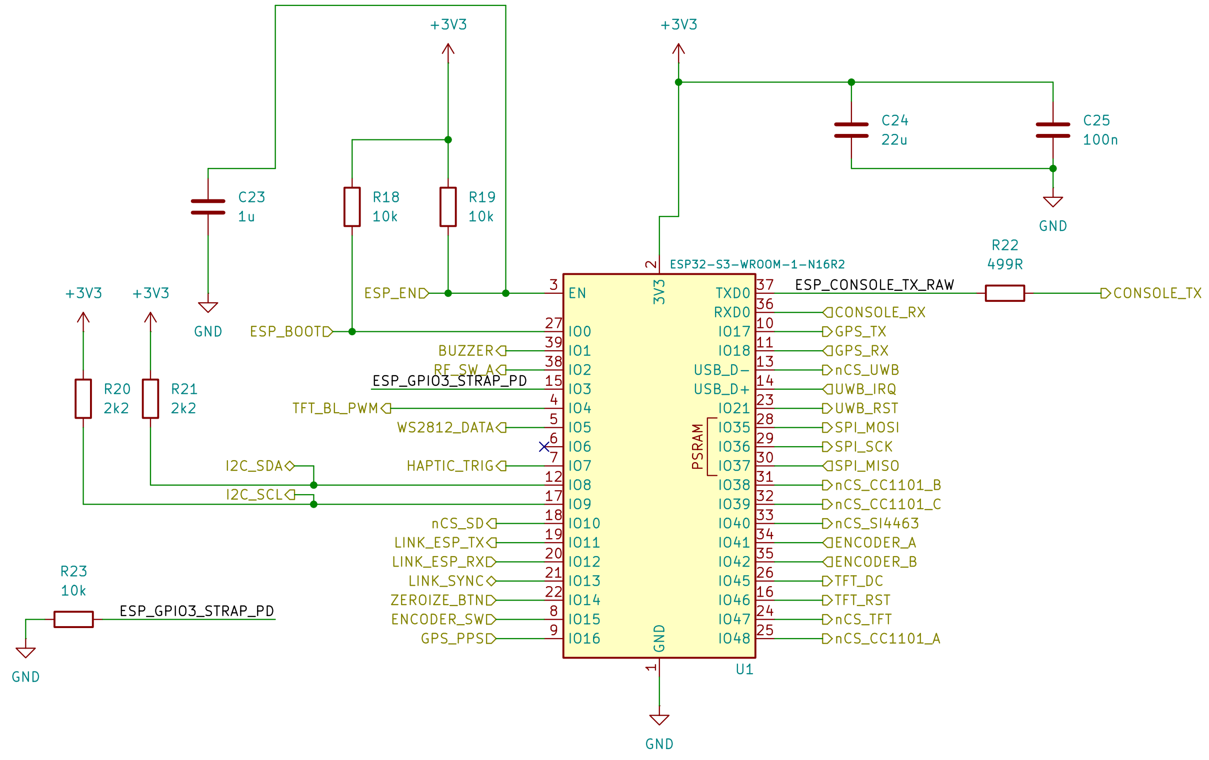
STRAPPING — GPIO0, GPIO45, and GPIO46 are sampled during reset. GPIO3 is also a strapping pin and remains unused. Connected peripherals must remain high-impedance during reset and must not override the required strap levels.

UART NET NAMES ARE FROM THE ESP32 PERSPECTIVE:
CONSOLE_TX = U1 TXD0 -> CP2102N RXD
CONSOLE_RX = CP2102N TXD -> U1 RXD0
GPS_TX/RX and LINK_ESP_TX/RX follow the same ESP32-side convention.

I2C — R20/R21 are the only populated SDA/SCL pull-ups. Do not duplicate I2C pull-ups on peripheral sheets.

BRING-UP — Provide accessible test pads/header access for +3V3, GND, ESP_EN, ESP_BOOT/GPIO0, CONSOLE_TX/RX, and LINK_ESP_TX/RX/SYNC.

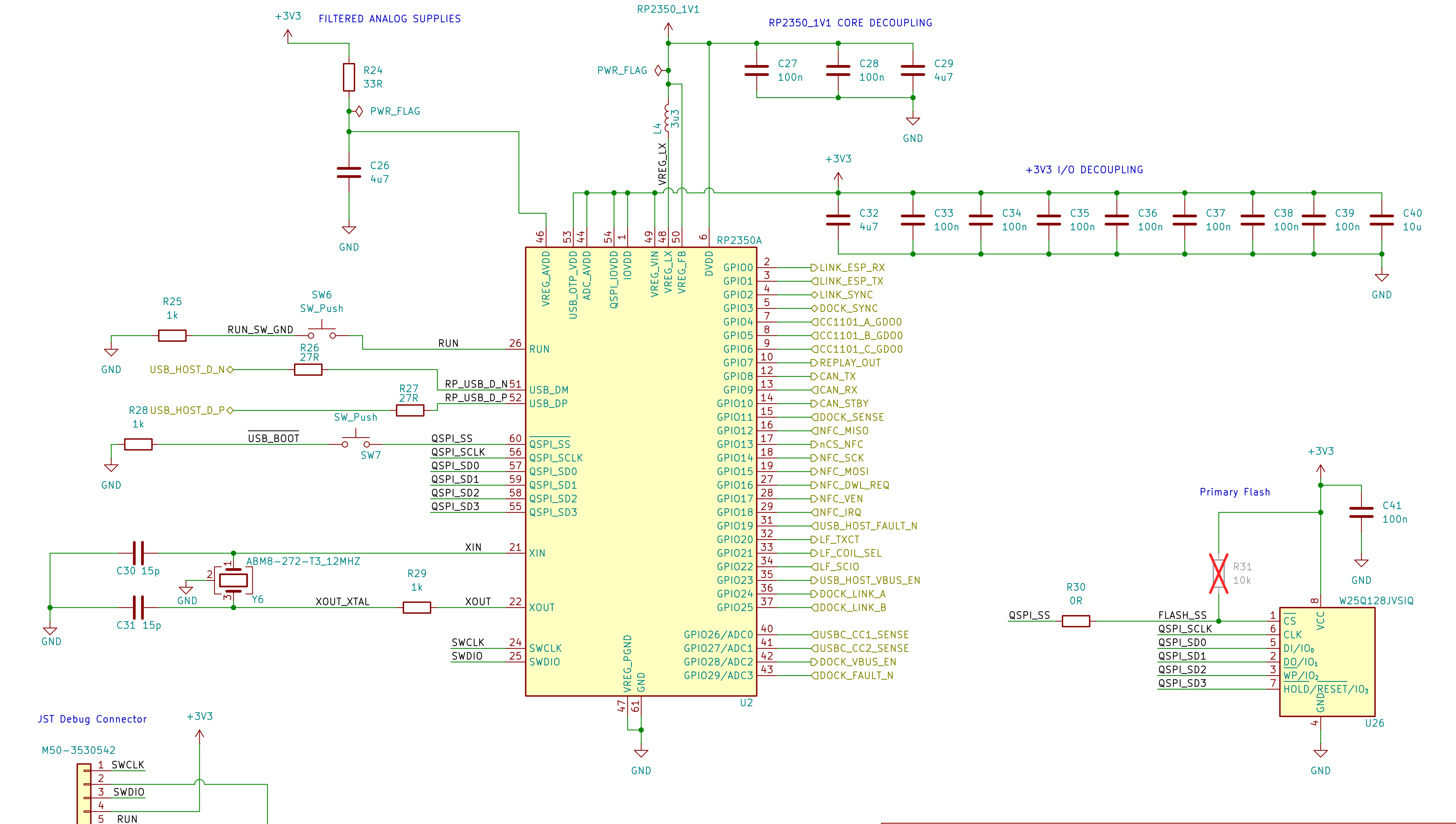
UART — Place 499R immediately beside U1 pin 37. Keep CONSOLE_TX/RX short, ground-referenced, and away from the antenna.



Sheet: /MAIN_CONTROLLER/
File: MAIN_CONTROLLER_sch.kicad_sch

Title: ESP32-S3 Main Controller and System Interfaces

Size: A4 | Date: | Rev: A
KiCad E.D.A. 10.0.5 | Id: 3/9



Sheet: /RP2350_COPROCESSOR/

File: RP2350_COPROCESSOR.kicad_sch

Title: RP2350 Coprocessor, Flash, Debug and USB Host

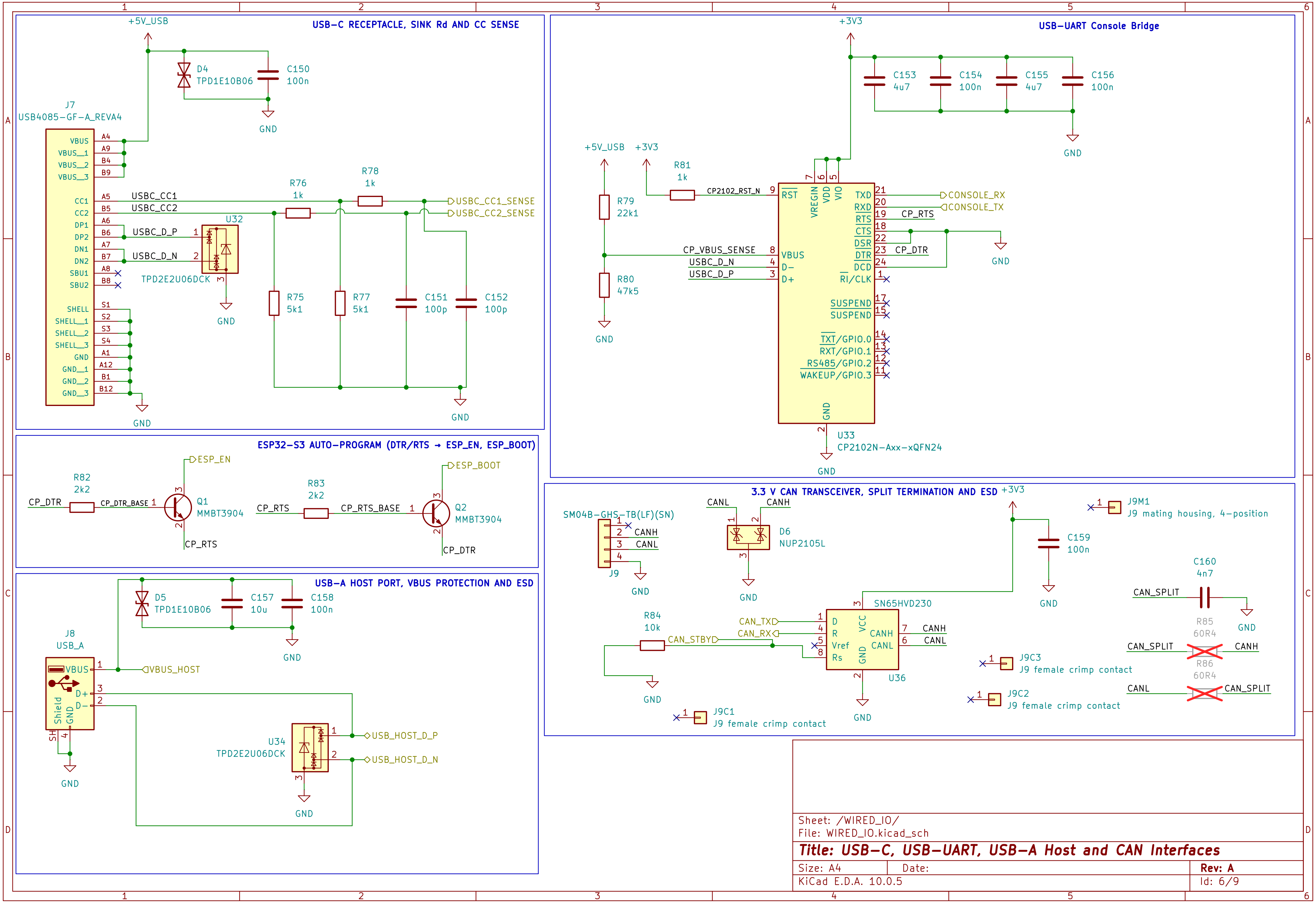
Size: A4 | Date: | Rev: A

KiCad E.D.A. 10.0.5 | Id: 4/9

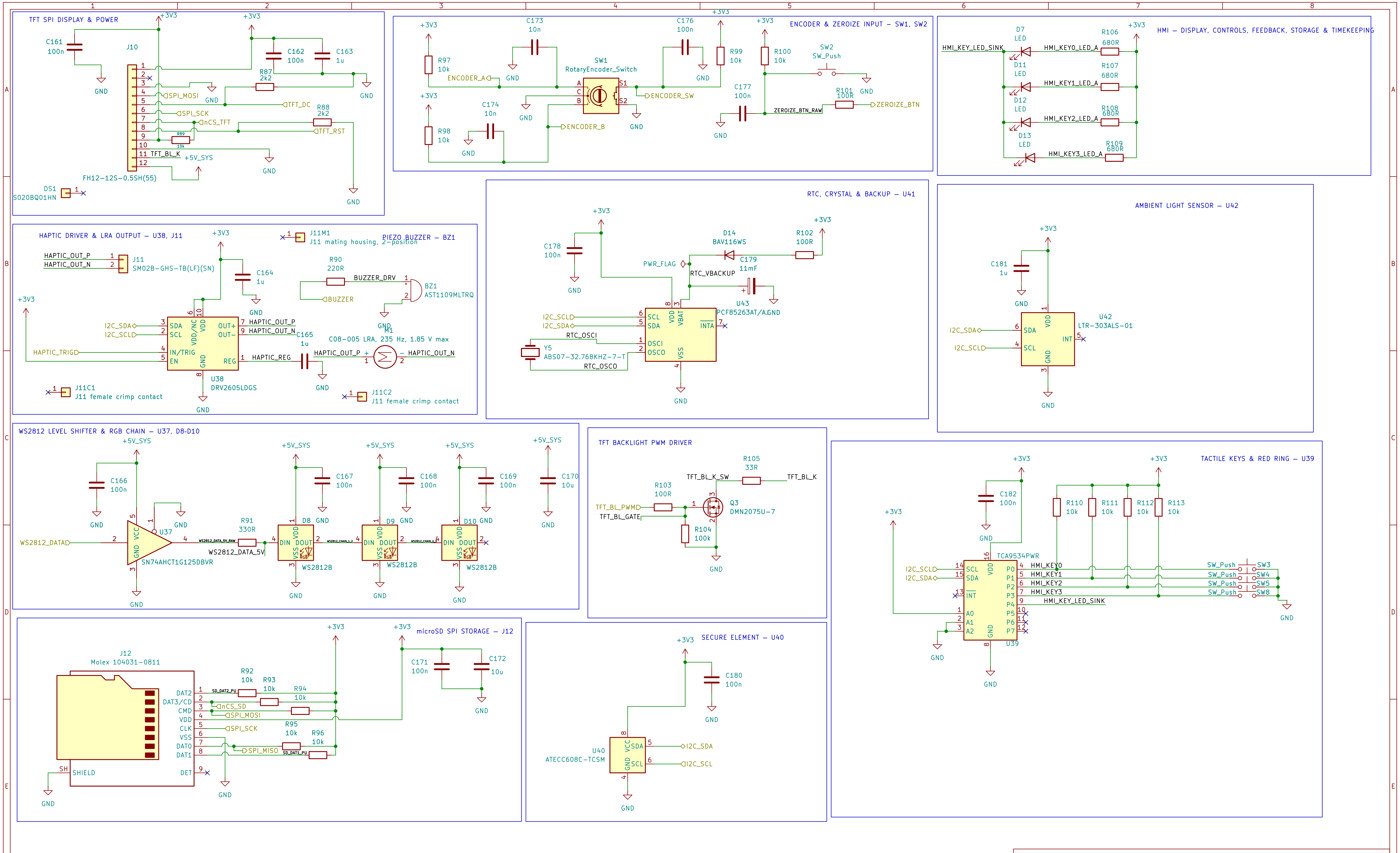
Sheet: /SUB_GHZ_RF/
File: SUB_GHZ_RF.kicad_sch

Title: Sub–GHz RF Transceivers, Front Ends and Band–Selectable

Size: A1 | Date: | Rev: A
KiCad E.D.A. 10.0.5 | Id: 5/9



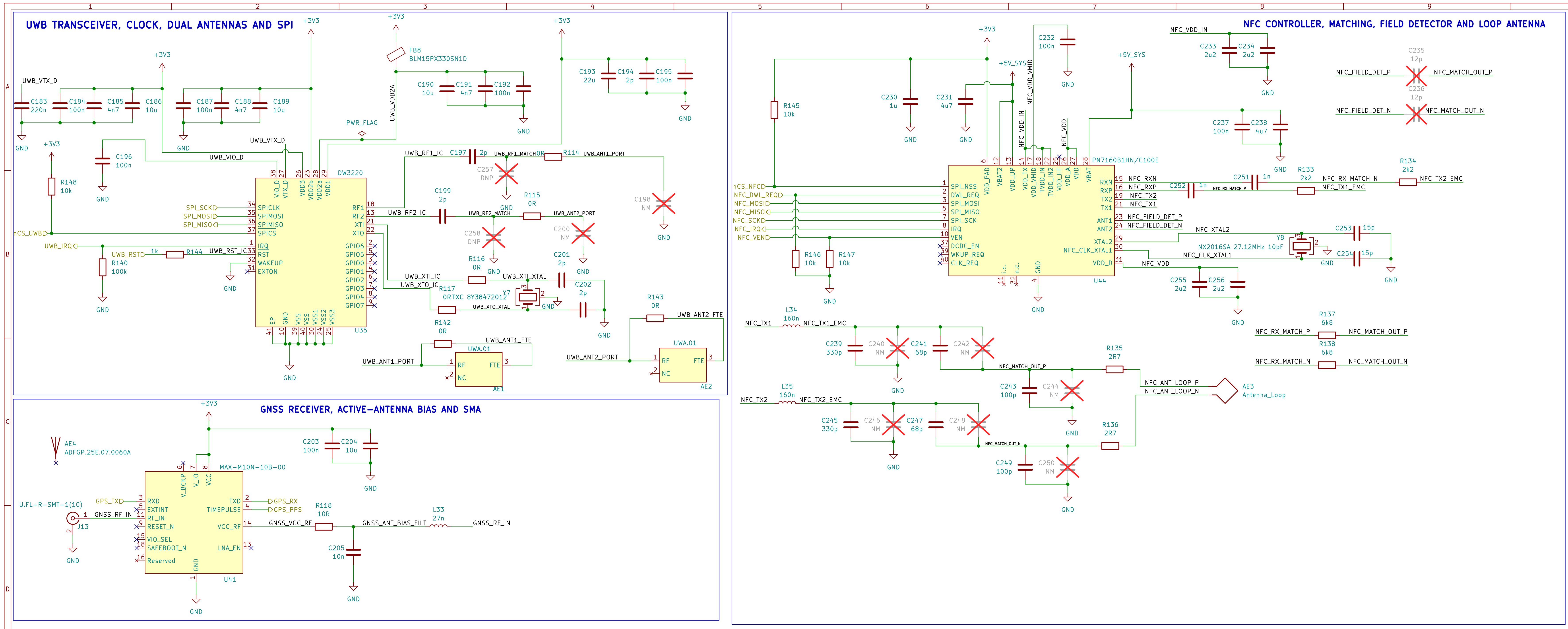
USB-C RECEPTACLE, SINK Rd AND CC SENSE
+5V_USB
D4
TPD1E10B06
C150
100n
GND
J7
USB4085-GF-A_REVA4
VBUS
VBUS__1
VBUS__2
VBUS__3
CC1
CC2
DP1
DP2
DN1
DN2
SBU1
SBU2
SHELL
SHELL__1
SHELL__2
SHELL__3
GND
GND__1
GND__2
GND__3
A4
A9
B4
B9
A5
B5
A6
B6
A7
B7
A8
B8
S1
S2
S3
S4
A1
A12
B1
B12
USBC_CC1
USBC_CC2
USBC_D_P
USBC_D_N
U32
TPD2E2U06DCK
R76
1k
R78
1k
R75
5k1
R77
5k1
C151
100p
C152
100p
USBC_CC1_SENSE
USBC_CC2_SENSE
USB-UART Console Bridge
+3V3
C153
4u7
C154
100n
C155
4u7
C156
100n
R79
22k1
R80
47k5
R81
1k
CP2102_RST_N
CP_VBUS_SENSE
USBC_D_N
USBC_D_P
RST
VBUS
D-
D+
VREGIN
VDD
VIO
TXD
RXD
RTS
CTS
DSR
DTR
DCD
RI/CLK
SUSPEND
SUSPEND
TXT/GPIO.0
RXT/GPIO.1
RS485/GPIO.2
WAKEUP/GPIO.3
CONSOLE_RX
CONSOLE_TX
CP_RTS
CP_DTR
U33
CP2102N-Axx-xQFN24
ESP32-S3 AUTO-PROGRAM (DTR/RTS → ESP_EN, ESP_BOOT)
R82
2k2
CP_DTR
CP_DTR_BASE
Q1
MMBT3904
ESP_EN
CP_RTS
R83
2k2
CP_RTS_BASE
Q2
MMBT3904
ESP_BOOT
3.3 V CAN TRANSCEIVER, SPLIT TERMINATION AND ESD
SM04B-GHS-TB(LF)(SN)
J9
CANH
CANL
D6
NUP2105L
J9M1
J9 mating housing, 4-position
C159
100n
C160
4n7
CAN_SPLIT
R85
60R4
R86
60R4
R84
10k
CAN_TX
CAN_RX
CAN_STBY
SN65HVD230
U36
D
R
Vref
Rs
VCC
CANH
CANL
J9C1
J9 female crimp contact
J9C2
J9 female crimp contact
J9C3
J9 female crimp contact
USB-A HOST PORT, VBUS PROTECTION AND ESD
D5
TPD1E10B06
C157
10u
C158
100n
J8
USB_A
VBUS
D+
D-
Shield
GND
VBUS_HOST
U34
TPD2E2U06DCK
USB_HOST_D_P
USB_HOST_D_N
Sheet: /WIRED_IO/
File: WIRED_IO.kicad_sch
Title: USB-C, USB-UART, USB-A Host and CAN Interfaces
Size: A4
Date:
Rev: A
KiCad E.D.A. 10.0.5
Id: 6/9

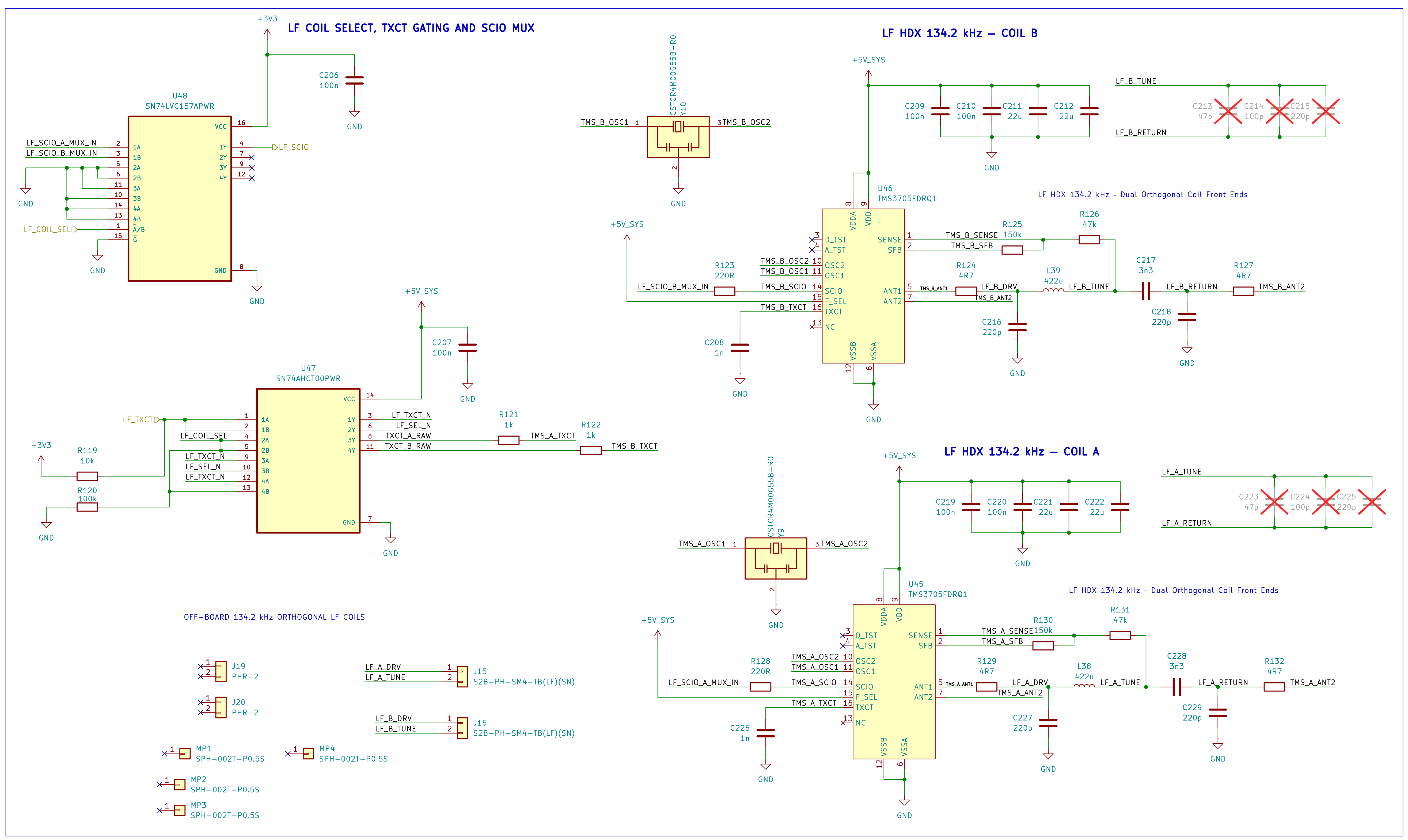


Sheet: /HMI/
File: HMI.kicad_sch

Title: Human-Machine Interface, Display, Controls And Storage

Size: A3	Date:	Rev: A
KiCad E.D.A. 10.0.5		Id: 7/9





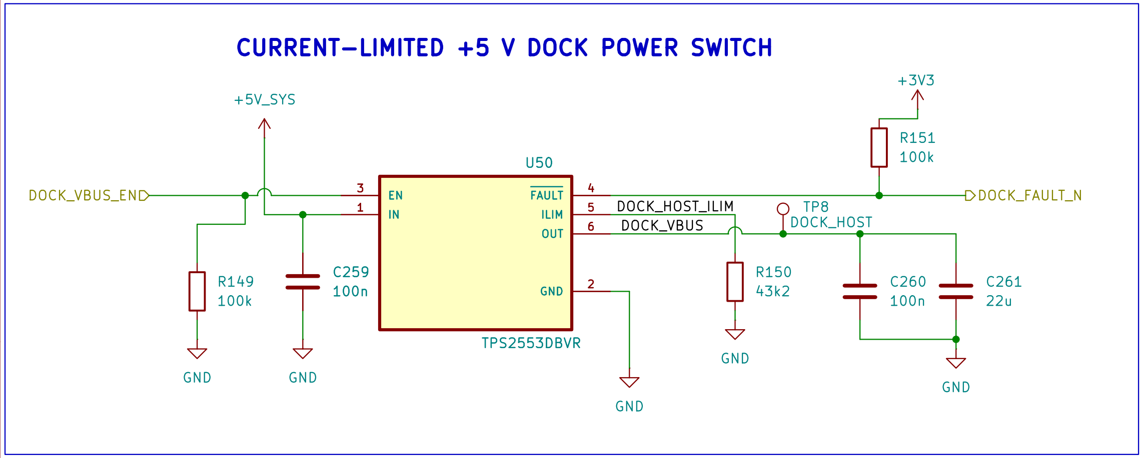
UWB PDoA ANTENNA LAYOUT:

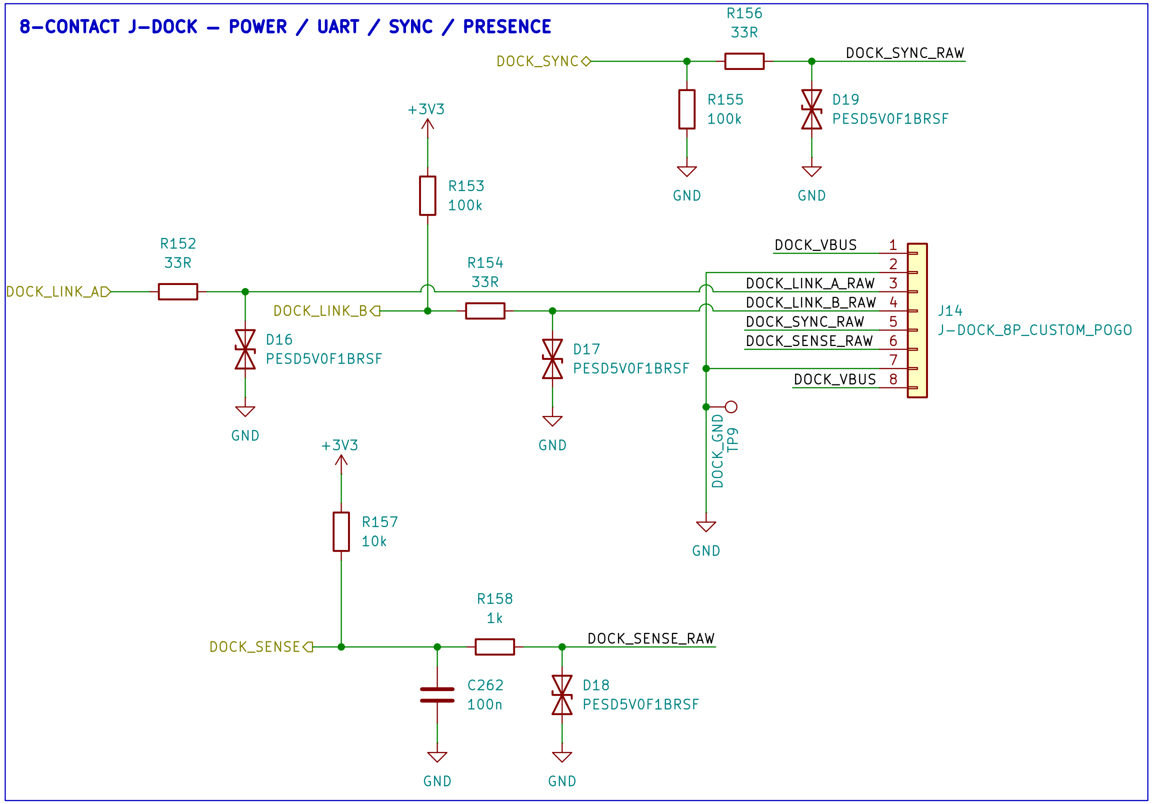
- Place AE1 and AE2 on the same PCB edge with identical orientation.
- Follow the Taoglas UWA.01 land pattern and maintain its 7.5 × 6.0 mm all-layer copper keepout for each antenna.
- Route RF1 and RF2 as phase-matched 50 Ω traces with identical transitions and grounding.
- Target antenna phase-centre spacing: 20.8 mm for channel 5 or 16.9 mm for channel 9 (0.45λ).
- Confirm the primary PDoA channel before placement.
- VNA-tune both networks and perform zero-angle PDoA calibration after assembly in the final enclosure.

Sheet: /POSITIONING/
File: POSITIONING.kicad_sch

Title: UWB, GNSS, NFC and LF Positioning / Credential Interfaces

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.5		Id: 8/9





J–DOCK CONTACT SEQUENCE:
pins 1/2/7/8 are long power/ground contacts; pins 3/4/5 are medium signal contacts;
pin 6 DOCK_SENSE is the shortest contact. DOCK_VBUS remains disabled until DOCK_SENSE is debounced LOW.
On removal, disable DOCK_VBUS when DOCK_SENSE becomes HIGH.
Dock UART operates at 921600 baud using the existing framed CRC protocol.

Sheet: /ANCHOR_DOCK/
File: ANCHOR_DOCK.kicad_sch

Title: BACK–FACE ANCHOR DOCK INTERFACE

Size: A4 Date: Rev: A
KiCad E.D.A. 10.0.5 Id: 9/9